Classification Models, Evaluation Metrics & Handling Imbalanced Data

Task 4 | Dataset: Breast Cancer Wisconsin | Models: Logistic Regression, Balanced Logistic Regression, Decision Tree

1. Metric Selection

1.1 Objective and Dataset

The objective was to compare classification models for distinguishing malignant and benign cases in the Breast Cancer Wisconsin dataset available through Scikit-learn. The dataset contains 569 samples and 30 numerical features. Label 0 represents malignant and label 1 represents benign. There are 212 malignant samples and 357 benign samples.

The data was divided into training and test sets using an 80:20 stratified split with `random_state=42`. The test set contains 114 samples: 42 malignant and 72 benign. Stratification preserves the approximate class proportions in each subset.

1.2 Why These Metrics Were Selected

Accuracy measures the proportion of all predictions that are correct: **(TP + TN) / (TP + TN + FP + FN)**. It gives an overall summary, but can hide weak performance on a less frequent class.

Precision (malignant) measures how many cases predicted as malignant are actually malignant: **TP / (TP + FP)**. It matters because false positives may lead to unnecessary concern or additional tests.

Recall (malignant) measures how many actual malignant cases are correctly identified: **TP / (TP + FN)**. This is especially important in a medical classification task because a false negative means a malignant case was predicted as benign.

F1-score (malignant) is the harmonic mean of precision and recall: **2 × (Precision × Recall) / (Precision + Recall)**. It gives a single measure that balances missed malignant cases against false alarms.

ROC-AUC summarizes how well a model separates classes across decision thresholds. A higher value generally indicates stronger class discrimination. The reported AUC uses `predict_proba(X)[:, 1]`; because label 1 is benign, the values below are for the benign class.

1.3 Confusion Matrix Interpretation

For standard Logistic Regression, the confusion matrix (rows = actual class; columns = predicted class; order = malignant, benign) was:

Actual / Predicted	Malignant	Benign
Malignant	41	1
Benign	1	71

The model correctly identified 41 of 42 malignant cases and 71 of 72 benign cases. One malignant case was predicted as benign, and one benign case was predicted as malignant. This illustrates why class-specific metrics and the confusion matrix are useful alongside accuracy.

2. Imbalance Handling

2.1 Understanding the Class Distribution

Class imbalance occurs when the target classes have different numbers of samples. This dataset contains 357 benign cases and 212 malignant cases, so benign is the majority class. Although the imbalance is moderate, a model could appear accurate while missing cases from the less frequent class. For this reason, the evaluation focuses on malignant precision, recall and F1-score as well as overall accuracy.

2.2 Method: Class-Weighted Logistic Regression

To test an imbalance-handling approach, a second Logistic Regression model was trained with `class_weight="balanced"`. This setting assigns class weights inversely related to class frequencies in the training data, increasing the influence of the less frequent class during training.

Class weighting does not add or remove training examples; it changes how the learning objective accounts for errors across classes. The standard Logistic Regression model was used as the baseline. Both Logistic Regression models used `StandardScaler`, fitted on training data only and then applied to the test data, avoiding leakage from the test set into the scaling step.

2.3 Results and Interpretation

Metric	Standard Logistic Regression	Balanced Logistic Regression
Accuracy	98.2%	95.6%
Malignant precision	97.6%	91.1%
Malignant recall	97.6%	97.6%
Malignant F1-score	97.6%	94.3%
ROC-AUC (benign)	0.995	0.995

Both Logistic Regression models identified 41 of the 42 malignant cases, so class weighting did not improve malignant recall on this test set. However, the balanced model classified four benign cases as malignant, compared with one for the standard model. This corresponds with lower malignant precision and lower overall accuracy for the balanced model.

The experiment shows that class weighting can change the balance between error types, but does not guarantee improvement on every metric. Other approaches—such as oversampling, undersampling or SMOTE—could be explored later, but were not used in this comparison. Any resampling should be applied only to training data, leaving test data untouched.

Important: The class-weighted model is not automatically the better choice just because the dataset is imbalanced. Its usefulness depends on the goal and the observed test results.

3. Final Model Decision

3.1 Model Comparison

Model	Accuracy	Precision (Malignant)	Recall (Malignant)	F1 (Malignant)	ROC-AUC (Benign)
Logistic Regression	0.982	0.976	0.976	0.976	0.995
Balanced Logistic Regression	0.956	0.911	0.976	0.943	0.995
Decision Tree	0.912	0.848	0.929	0.886	0.916

3.2 Selected Model: Logistic Regression

Standard Logistic Regression was selected as the final model for this experiment because it delivered the strongest overall performance on the reported test set. It achieved 98.2% accuracy, 0.976 malignant precision, 0.976 malignant recall, 0.976 malignant F1-score and 0.995 ROC-AUC for the benign class.

The model correctly identified 41 of 42 malignant cases while producing only one false positive malignant prediction among the 72 benign cases. Compared with Balanced Logistic Regression, it matched malignant recall and ROC-AUC while achieving better accuracy, precision and F1-score. It also outperformed the Decision Tree on all metrics shown in the comparison table.

3.3 Limitations and Future Improvements

The results come from one stratified train-test split, and the test set includes only 42 malignant cases. Performance may vary with another split or a new dataset; the reported scores should not be treated as proof of clinical reliability. Future work could include stratified cross-validation, hyperparameter tuning, additional imbalance-handling methods, threshold analysis and evaluation on an independent dataset.

This is an educational machine learning experiment, not a clinically validated diagnostic system. It should not be used to make medical decisions without appropriate validation and expert oversight.

3.4 Conclusion

The experiment compared Logistic Regression, class-weighted Logistic Regression and a Decision Tree using multiple evaluation metrics. Standard Logistic Regression was selected because it achieved the strongest overall results in this comparison. The imbalance-handling experiment also showed that class weighting should be judged by its measured effect on the target metrics, not assumed to improve performance automatically.

Overall, the task demonstrates that sound model selection requires more than accuracy: it also requires examining class-specific errors, understanding the consequences of false negatives and false positives, and evaluating models using metrics aligned with the problem.